

Albumin

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This test is used for evaluation of nutritional status and blood oncotic pressure, and evaluation of renal disease with proteinuria and other chronic diseases.

High albumin may indicate dehydration. Look for increase in hemoglobin, hematocrit in such patients.

Low albumin is found with use of I.V. fluids, rapid hydration, overhydration; cirrhosis, other liver disease, including chronic alcoholism; in pregnancy and with oral contraceptive use; many chronic diseases including the nephrotic syndromes, neoplasia, protein-losing enteropathies (including Crohn's disease and ulcerative colitis), peptic ulcer, thyroid disease, burns, severe skin disease, prolonged immobilization, heart failure, chronic inflammatory diseases such as the collagen diseases and other chronic catabolic states.

Starvation, malabsorption, or malnutrition: In the absence of I.V. fluid therapy and in patients without liver or renal disease, low albumin may be regarded as an indication of inadequate body protein reserves. It is described as the most common nutrition-related abnormality in patients with infection.¹ Serum albumin has a half-life of about 18 to 20 days. Its half-life is decreased in patients with catabolic states: infection and with protein loss through the kidneys (eg, nephrosis), gastrointestinal tract, and skin (eg, burns). Its prognostic application is most useful in patients with weight loss, anorexia, stress, surgical therapy, hemorrhage, and infection. Total iron binding capacity $<240\text{ }\mu\text{g/dL}$ ¹ and/or low transferrin levels would support an impression of inadequate protein reserves. Absolute lymphocyte counts $<1500/\text{mm}^3$ may also be seen with protein malnutrition.² In severe malnutrition, albumin has been reported as $<2.5\text{ g/dL}$, total lymphocytes as $<800/\text{mm}^3$ and TIBC as $<150\text{ }\mu\text{g/dL}$.²

Albumin levels $\leq 2.0\text{--}2.5\text{ g/dL}$ may be the cause of edema (eg, nephrotic syndrome, protein-losing enteropathies).

Albumin, prealbumin, and transferrin are regarded as “negative” acute phase reactants (ie, these proteins decrease with acute inflammatory/infectious processes).

Low albumin values are associated with longer hospital stay.¹

Custom Additional Information

Twenty-four hour urine collection to measure protein loss is helpful in work-up of some patients with hypoalbuminemia. Other tests **useful in assessment of nutritional status** include TIBC, transferrin, iron, absolute lymphocyte count, and vitamin B₁₂/folate levels.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Male (g/dL)	Female (g/dL)
0 to 7 d	3.6–4.9	3.6–4.9
8 to 30 d	3.5–4.6	3.5–4.6
1 to 6 m	3.7–4.8	3.7–4.8
7 m to 2 y	4.0–5.0	4.0–5.0
3 to 5 y	4.1–5.0	4.1–5.0
6 to 12 y	4.2–5.0	4.2–5.0
13 to 30 y	4.3–5.2	4.0–5.0
31 to 50 y	4.1–5.1	3.9–4.9
51 to 60 y	3.8–4.9	3.8–4.9
61 to 70 y	3.9–4.9	3.9–4.9
71 to 80 y	3.8–4.8	3.8–4.8
81 to 89 y	3.7–4.7	3.7–4.7
>89 y	3.6–4.6	3.6–4.6

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Fluoride plasma specimen; gross hemolysis; lipemia; improper labeling

References

Tonks DB. A study of the accuracy and precision of clinical chemistry determinations in 170 Canadian laboratories. *Clin Chem*. 1963 Apr; 9:217-233. PubMed 13985504 (<http://www.ncbi.nlm.nih.gov/pubmed/13985504>)

Footnotes

1. Anderson CF, Wochos DN. The utility of serum albumin values in the nutritional assessment of hospitalized patients. *Mayo Clin Proc*. 1982 Mar; 57(3):181-184. PubMed 6801397 (<http://www.ncbi.nlm.nih.gov/pubmed/6801397>)
2. Shapiro M, Rhodes JB, Beyer PL. Malnutrition. Recognition and correction by enteral nutrition. *J Kans Med Soc*. 1983 Jun; 84(6):341-345, 356. PubMed 6409975 (<http://www.ncbi.nlm.nih.gov/pubmed/6409975>)